(f) Let V be a representation of H. Prove that $\operatorname{Ind}_H^G(V^*) \cong \left(\operatorname{Ind}_H^G V\right)^*$ as representations of G. [Hint: Write $\operatorname{Ind}_H^G V$ as $k[G]_2 \otimes_{k[H]} V$ and write $\operatorname{Ind}_H^G(V^*)$ as $\operatorname{Hom}_{k[H]}(k[G]_1, V^*)$. Prove that the map $\operatorname{Hom}_{k[H]}(k[G]_1, V^*) \times \left(\operatorname{Ind}_H^G(V)\right) \to k$ given by $\left(f, \left(x \otimes_{k[H]} v\right)\right) \mapsto (f(Sx))(v)$ is a nondegenerate G-invariant bilinear form, where $S : k[G] \to k[G]$ is the linear map defined by $Sg = g^{-1}$ for every $g \in G$.]

5.11. Examples

Here are some examples of induced representations (we use the notation for representations from the character tables).

(1) Let $G = S_3$, $H = \mathbb{Z}_2$. Using the Frobenius reciprocity, we obtain $\operatorname{Ind}_H^G \mathbb{C}_+ = \mathbb{C}^2 \oplus \mathbb{C}_+$ and $\operatorname{Ind}_H^G \mathbb{C}_- = \mathbb{C}^2 \oplus \mathbb{C}_-$.

(2) Let $G = S_3$, $H = \mathbb{Z}_3$. Then we obtain $\operatorname{Ind}_H^G \mathbb{C}_+ = \mathbb{C}_+ \oplus \mathbb{C}_-$, $\operatorname{Ind}_H^G \mathbb{C}_\epsilon = \operatorname{Ind}_H^G \mathbb{C}_{\epsilon^2} = \mathbb{C}^2$.

(3) Let $G = S_4$, $H = S_3$. Then $\operatorname{Ind}_H^G \mathbb{C}_+ = \mathbb{C}_+ \oplus \mathbb{C}_-^3$, $\operatorname{Ind}_H^G \mathbb{C}_- = \mathbb{C}_- \oplus \mathbb{C}_+^3$, $\operatorname{Ind}_H^G \mathbb{C}^2 = \mathbb{C}^2 \oplus \mathbb{C}_-^3 \oplus \mathbb{C}_+^3$.

Problem 5.11.1. Compute the decomposition into irreducibles of all the representations of A_5 induced from the irreducible representations of

(a) $\mathbb{Z}_2$;

(b) $\mathbb{Z}_3$;

(c) $\mathbb{Z}_5$;

(d) A_4;

(e) $\mathbb{Z}_2 \times \mathbb{Z}_2$.

5.12. Representations of S_n

In this subsection we give a description of the representations of the symmetric group S_n for any n.

Definition 5.12.1. A **partition** λ of n is a representation of n in the form $n = \lambda_1 + \lambda_2 + \cdots + \lambda_p$, where λ_i are positive integers and $\lambda_i \geq \lambda_{i+1}$.